REAPER-Notator v2.50.4

Comprehensive Codebase Volume & Structural Analysis

TOTAL LINES	EXECUTABLE CODE	SOURCE FILES	PATTERN LIBRARY
79,305	67,633	92 Lua	1,500 JSON

1. Architecture & Module Volume Breakdown

Module Directory	Files	Total Lines	Code Lines	Comments	Blank	Code Ratio
modules/services	37	41,319	35,490	2,492	3,533	85.9%
modules/rendering	7	14,349	12,349	848	1,199	86.1%
modules/ui	25	13,441	11,005	965	1,498	81.9%
tests	2	3,307	2,933	175	233	88.7%
modules	3	2,638	2,323	481	162	88.1%
modules/input	2	2,123	1,853	92	180	87.3%
Root	1	1,338	1,067	167	108	79.7%
modules/classes	15	790	613	110	96	77.6%
TOTAL	**92**	**79,305**	**67,633**	**5,330**	**7,009**	**85.3%**

2. Key Core System Files by Volume (Top 15)

File Name	Subsystem / Module	Lines	Code	Comments
midi_service.lua	modules/services	7,530	6,653	288
score_canvas.lua	modules/rendering	5,205	4,550	246
engraver.lua	modules/rendering	4,389	3,800	300
record_quantize_service.lua	modules/services	2,856	2,459	193
slur_service.lua	modules/services	2,656	2,316	125
canvas_context_menus.lua	modules/rendering	2,339	2,045	105
musicxml_import_service.lua	modules/services	2,193	1,911	125
test_all_modules.lua	tests	1,989	1,683	149
musicxml_export_service.lua	modules/services	1,715	1,452	145
melody_pathfinder_service.lua	modules/services	1,643	1,375	117
portamento_service.lua	modules/services	1,561	1,322	90
glissando_service.lua	modules/services	1,489	1,257	82
midi_editor_drawer.lua	modules/ui	1,449	1,206	118
state.lua	modules	1,432	1,252	245
settings_modal.lua	modules/ui	1,420	1,154	96

3. Curated Orchestral Pattern Library (1,500 Patterns)

Category Folder	Genre / Instrumentation Scope	Pattern Count
01_Strings_Staccato	Cinematic Strings (Staccato / Spiccato Ostinatos)	150 patterns
02_Strings_Pizzicato	Orchestral Strings (Pizzicato / Bartók Snap)	150 patterns
03_Brass_Blockbuster	Epic Brass (Horns, Trumpets, Trombones, Tubas)	150 patterns
04_Cinematic_Melodies	Heroic Themes & Symphonic Lead Melodies	150 patterns
05_Counter_Melodies	Polyphonic Counter-Melodies & Descant Voices	150 patterns
06_Woodwinds_Textures	Woodwind Runs, Flutter-Tongue & Textures	150 patterns
07_Cinematic_Piano	Concert Grand Piano (Arpeggios & Ostinatos)	150 patterns
08_Ancient_Harp_Greek_Roman	Antiquity Concert Harp (Greek & Roman Hymns)	150 patterns
09_Orchestral_Percussion	Orchestral & Cinematic Percussion (Timpani, Taiko, Drums)	150 patterns
10_Low_Strings_Celli_Basses	Low Strings Tension & Rhythm (Staccato Celli & Double Basses)	150 patterns
TOTAL PATTERNS	**10 Curated Production Categories**	**1,500 patterns**

4. Specialized Orchestral & Engraving Services Subsystem

Service Subsystem File	Musical & Architectural Responsibility	Code Lines
annotation_service.lua	Annotation Layer Engine - Per-track multi-layer free	972
circle_of_fifths_service.lua	Circle of Fifths & Harmonic Palette - Chord wheel, t	680
melody_pathfinder_service.lua	Melodic Pathfinder Engine - Corpus-trained Viterbi i	1,375
melody_style_corpus.lua	Composer Style Corpus - Statistical Markov transitio	248
divisi_service.lua	Divisi Engine - Split polyphonic voices onto full du	434
divisio_vocum_service.lua	Divisio Vocum - Orchestral Voice Division across tra	308
dynamics_engine.lua	Orchestral Dynamics Engine - Bezier/Linear CC automa	1,128
filter_notes_service.lua	Filter Irregular Notes - Metric onset analysis and v	308
frame_timer_service.lua	Native High-Refresh Timer - 1ms kernel multimedia ti	147
glissando_service.lua	Glissando Engine - Smooth chromatic / diatonic slide	1,257
imgui_guard_service.lua	ImGui Stack Guard Engine - Zero-leak Begin/End & Beg	358
key_signature_service.lua	Key Signature Service - Project/Take key signatures	1,019
midi_service.lua	Core MIDI Engine - Low-latency take I/O and event sy	6,653
musicxml_export_service.lua	MusicXML 4.0 Exporter - Standard score interchange f	1,452
musicxml_import_service.lua	MusicXML 4.0 Importer - Orchestral part loading and	1,911
audio_preview.lua	Audition Preview Engine - Non-destructive note click	1,062
reaticulate_parser.lua	Reaticulate Bridge - FX Slot 1 setup, bank sync & ke	876
record_quantize_service.lua	Record & Auto-Quantize Engine - Live MIDI capture, S	2,459
scale_service.lua	Scale & Harmony Service - Chord Track harmonic trans	925
slur_service.lua	Slur Engine - Ergonomic Bezier phrasing curves and k	2,316
portamento_service.lua	Portamento Service - High-resolution micro-tonal pit	1,322
pattern_service.lua	Pattern Engine - Orchestral motif browser and timeli	582
rehearsal_mark_service.lua	Rehearsal Marks - Custom/auto structural navigation	368
fermata_service.lua	Fermata Service - Score-wide pause holds and tempo m	232
pedal_service.lua	Pedal Service - Classical damper and sustain pedal C	530

Quality Assurance & Standards Status: 100% Certified

All 92 Lua files, 159 unit tests, and 1,500 pattern JSON files verified with 0 errors. Fully compliant with Gardner Read & Elaine Gould engraving standards.